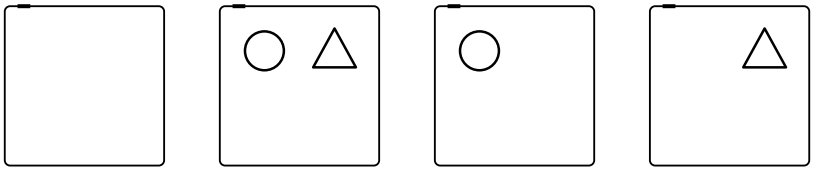
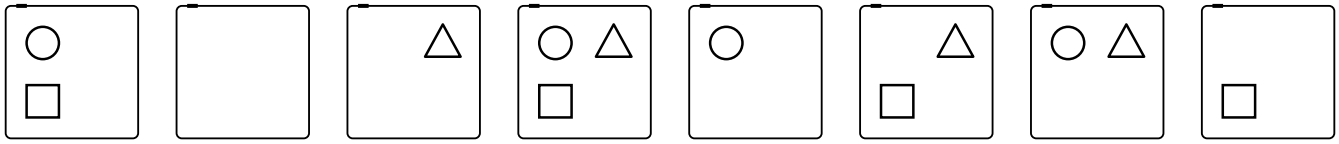
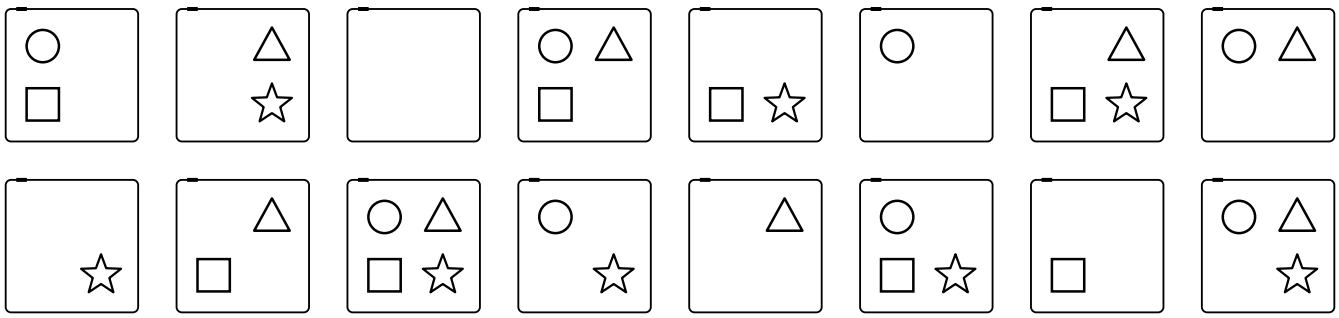
A card fits inside another when all its pictures are on the other card. The empty card fits inside every card. In a collection, no card may fit inside another. Use a card only once in a collection.

Problem 1: For each deck, find the largest collection you can. Explain why no larger collection can work.

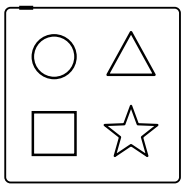


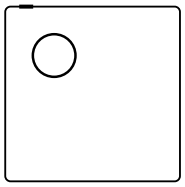


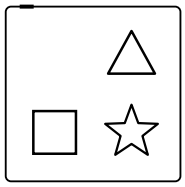
Problem 2: Use this sixteen-card deck. Find the largest collection you can, and decide whether you can rule out every larger collection.

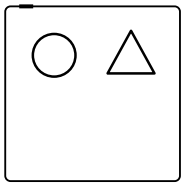


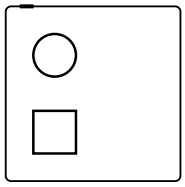
Problem 3: Use the sixteen-card deck. Keep each starting collection and add as many cards as possible. Explain whether a collection that cannot take another card must be as large as possible.



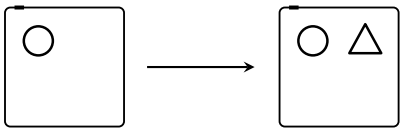




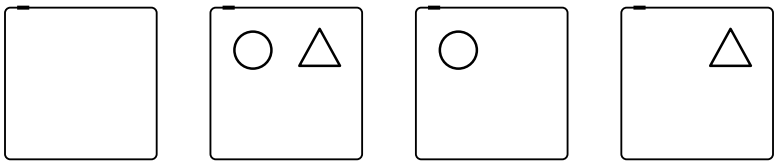


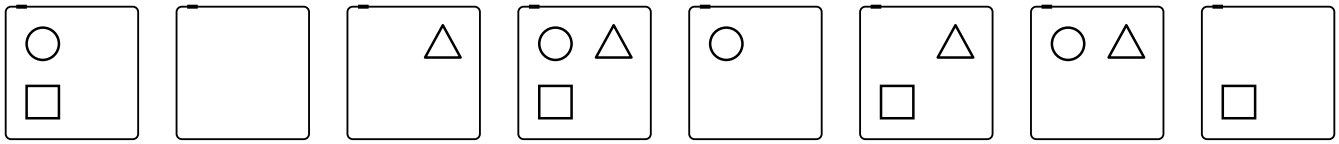


Problem 4: For each deck, arrange every card into exactly one row, with each card fitting inside the next card in its row. A card can be alone in a row. Make as few rows as possible.



One nested row.

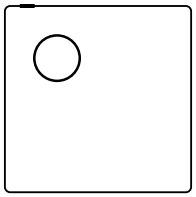




Problem 5: Arrange all sixteen cards from Problem 2 into rows, with each card fitting inside the next card in its row. Use every card exactly once and make as few rows as possible.

Problem 6: What is the largest possible collection from the sixteen-card deck? Give a collection of that size and explain why every larger collection breaks the rule.

Problem 7: Use the sixteen-card deck. Find the largest collection that includes the card shown. Explain why no larger collection can include it.



Problem 8: Find every collection that reaches your largest size from Problem 6. Explain why your list is complete.